

1. Database Design

Overview

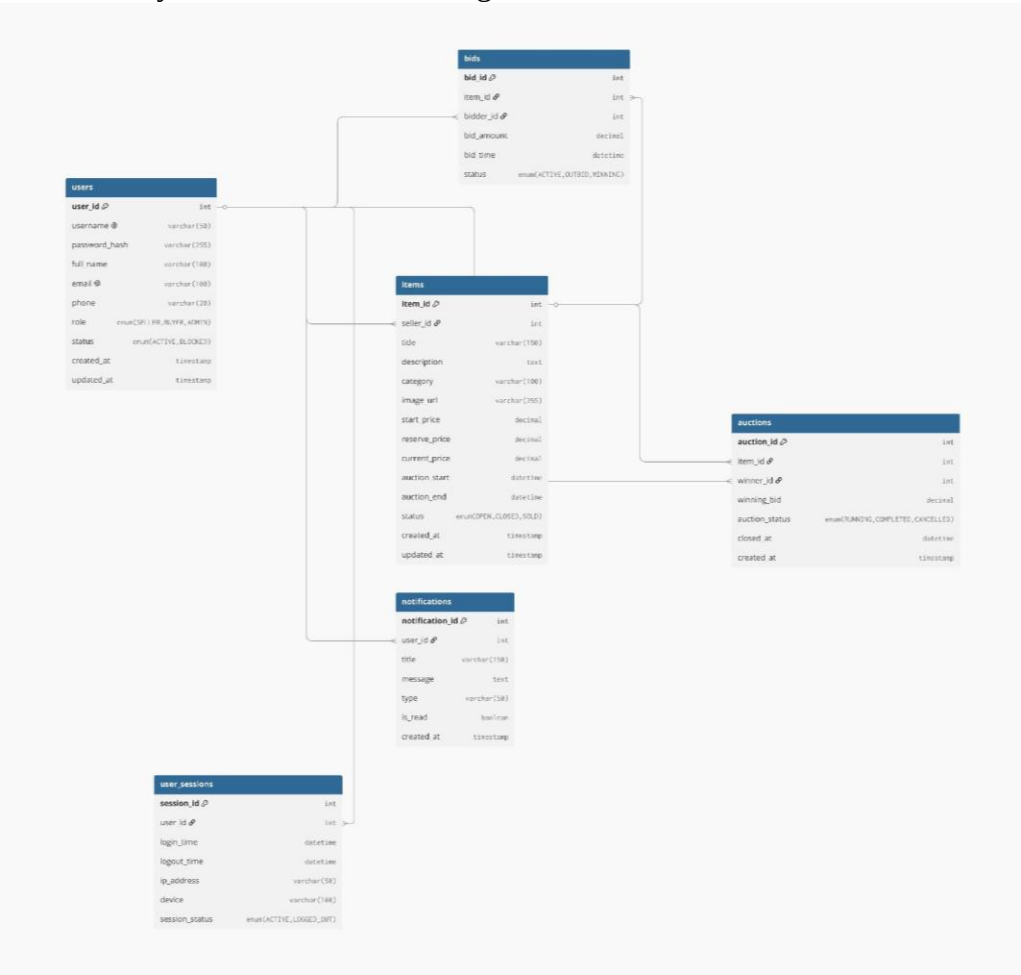
The database is the core component of the Online Auction System, responsible for storing and managing all essential information. It maintains records of users, auction items, bids, auction results, notifications, and user login sessions. By organizing data efficiently, the database ensures secure storage and smooth operation of all auction-related activities.

Main Tables

- **Users:** Stores information about buyers and sellers.
- **Items:** Contains details of products listed for auction.
- **Bids:** Records all bids submitted by buyers.
- **Auctions:** Maintains auction status, results, and winner information.
- **Notifications:** Stores messages and alerts sent to users.
- **User Sessions:** Keeps track of user login and logout activities.

Relationships

- A single seller can list multiple items for auction.
- A single buyer can place multiple bids on different items.
- Each auction item can receive multiple bids from different buyers.
- Every auction has one winning bidder.



2. System Architecture

Overview

The Online Auction System follows a Client-Server Architecture. Buyers and sellers use the client application to connect to the Auction Server through TCP sockets. The Auction Server processes all requests and communicates with the Storage Server, which accesses the database.

Components

Client

The client is used by buyers and sellers to log in, upload items, search auctions, place bids, and view results.

Auction Server

The Auction Server handles user authentication, manages auctions, processes bids, selects the winner, and sends notifications.

Storage Server

The Storage Server manages all database operations such as saving, updating, and retrieving data.

Database

The database stores all system information, including users, items, bids, auctions, notifications, and user sessions.

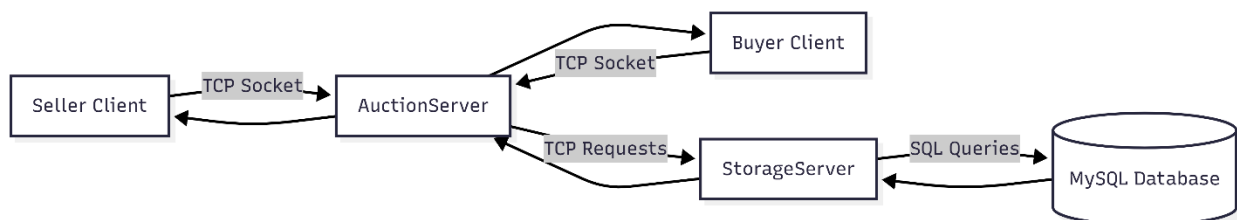
Working Flow

1. User sends a request from the client.
2. Auction Server processes the request.
3. Storage Server accesses the database if needed.
4. Database returns the required data.
5. Auction Server sends the response back to the user.

Benefits

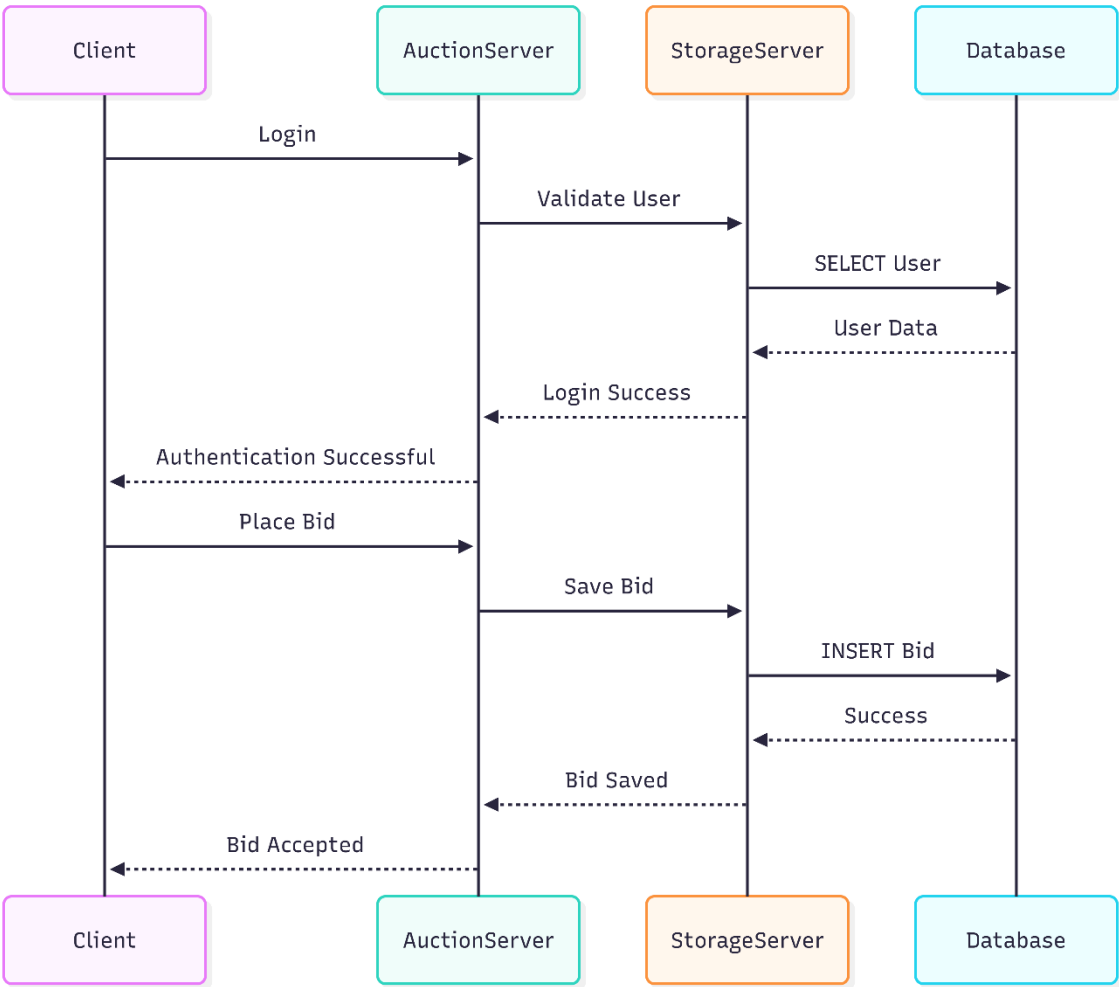
- Easy to manage and maintain.
- Secure data storage.
- Fast communication using TCP.
- Supports multiple users and future expansion.

These explanations are concise, easy to understand, and suitable for a project report.



This diagram explains the internal modules of the Auction Server. The Authentication module verifies users before allowing access. The Session Manager maintains active user sessions. Item Management handles auction items, Search Service finds available products, Bid Manager processes bids, Winner Selection determines the highest bidder, and Notification Service

informs buyers and sellers. All data is stored through the Storage Server in the database.



This sequence diagram shows how the system processes a user request. The client sends a request to the Auction Server. The server communicates with the Storage Server, which retrieves or updates information in the database. After processing the request, the response is sent back to the client. The same communication process is used for login, searching items, placing bids, and viewing auction results.

3. Business Logic

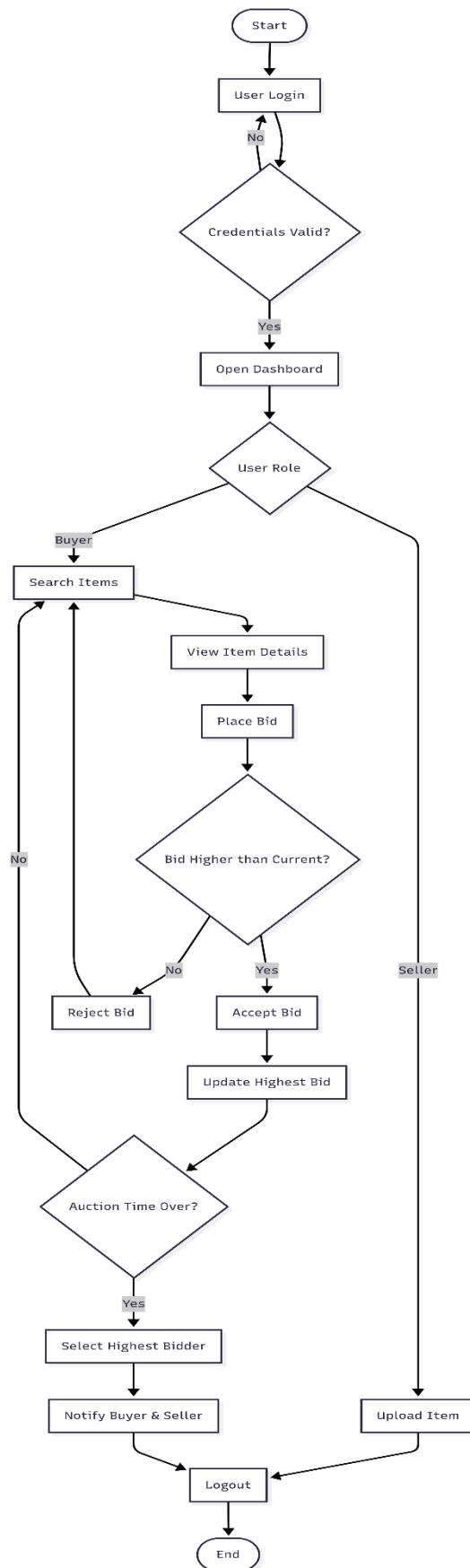
3.1 Business Workflow

3.2 Business Logic Explanation

The Online Auction System follows a simple workflow:

1. The user logs in using a valid username and password.
2. After successful login, the user accesses the dashboard.
3. Sellers can upload new auction items.
4. Buyers can search and view available auction items.
5. Buyers place bids on selected items.
6. The system checks whether the new bid is higher than the current highest bid.
7. If valid, the bid is accepted and stored; otherwise, it is rejected.
8. When the auction time ends, the system selects the highest bidder as the winner.

Rule	Description
User Login	Users must log in before accessing the system.
Authentication	Only valid users can perform auction operations.
Seller Access	Sellers can upload and manage auction items.
Buyer Access	Buyers can search items and place bids.
Bid Validation	Every new bid must be higher than the current highest bid.
Auction Closing	No bids are accepted after the auction end time.
Winner Selection	The highest valid bid wins the auction.
Notification	The system notifies both the seller and the winning buyer.
Logout	Users should log out after completing their tasks.



9. Both the seller and the winning buyer are notified.
10. The user logs out after completing the operations.

3.3 Business Rules

3.4 Main Modules

Authentication: Verifies user credentials and allows access only to registered users.

Item Management: Allows sellers to upload, update, and manage auction items.

Search Service: Enables buyers to search and view available auction items.

Bid Management: Receives bids, validates them, and updates the highest bid.

Auction Engine: Monitors auction time and automatically closes auctions when the time expires.

Winner Selection: Determines the highest bidder after the auction ends.

Notification Service: Sends auction results and winner information to the buyer and seller.

3.5 Advantages of the Business Logic

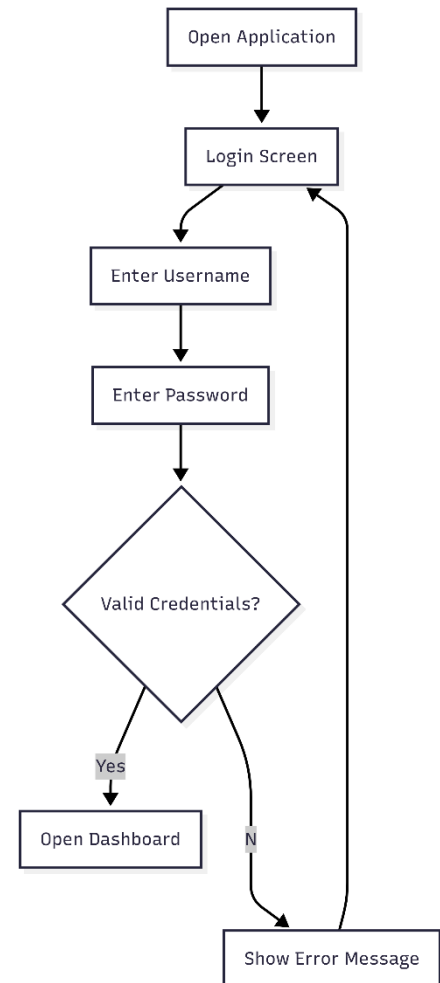
- Ensures only authenticated users access the system.
- Maintains fair bidding through bid validation.
- Automatically determines the auction winner.
- Separates responsibilities into different modules.
- Makes the system easy to maintain and extend.

4. Storyboards

Storyboard 1 – Login Screen

Explanation

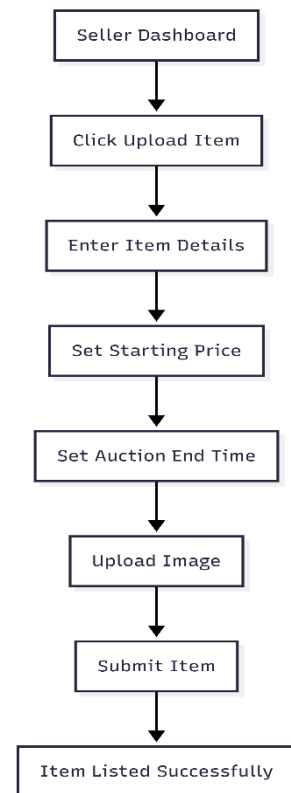
- User opens the Online Auction System.
- User enters a username and password.
- The system verifies the login credentials.
- If the credentials are correct, the user is redirected to the dashboard.
- If the credentials are incorrect, an error message is displayed.
- The user can try logging in again.



Storyboard 2 – Seller Upload Item

Explanation

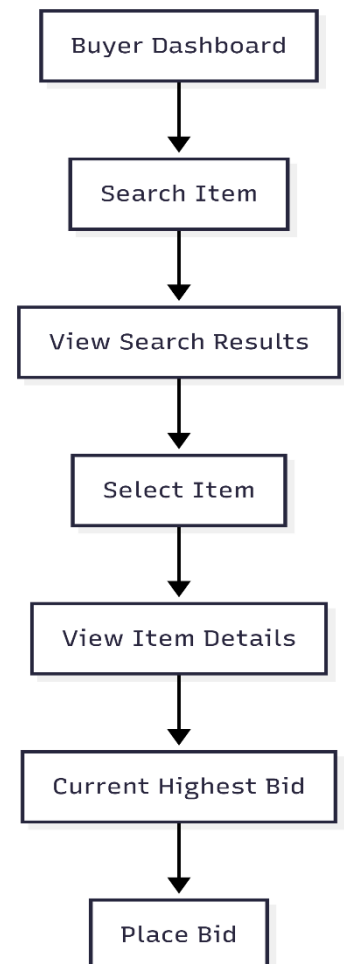
- The seller logs into the system.
- The seller opens the **Upload Item** page.
- Product details such as title, description, and category are entered.
- The seller sets the starting price and auction end time.
- An image of the item is uploaded.
- The item is submitted and becomes available for bidding.



4.3 Buyer Search & View Item

Explanation

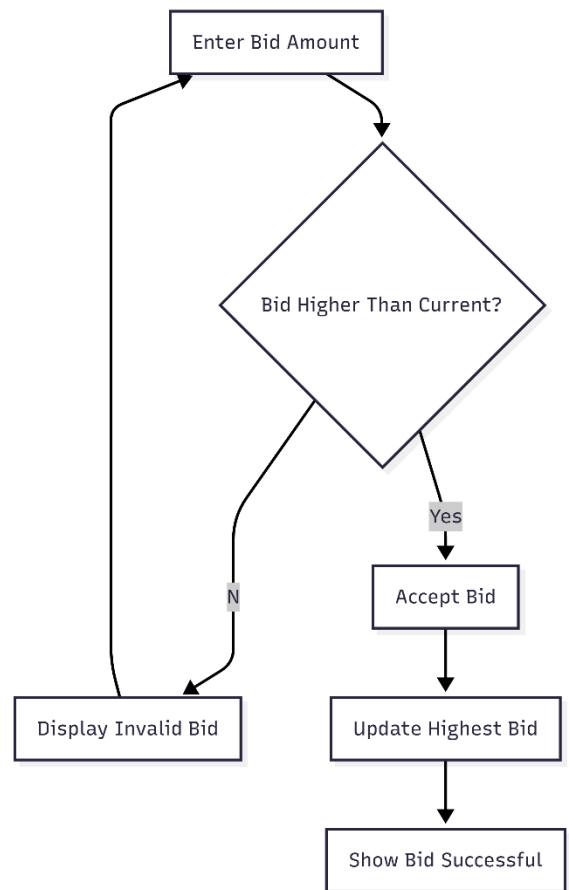
- The buyer logs into the system.
- The buyer searches for auction items.
- The system displays matching products.
- The buyer selects an item from the list.
- The system shows the item's details.
- The buyer views the current highest bid before placing a bid.



4.4 Place Bid

Explanation

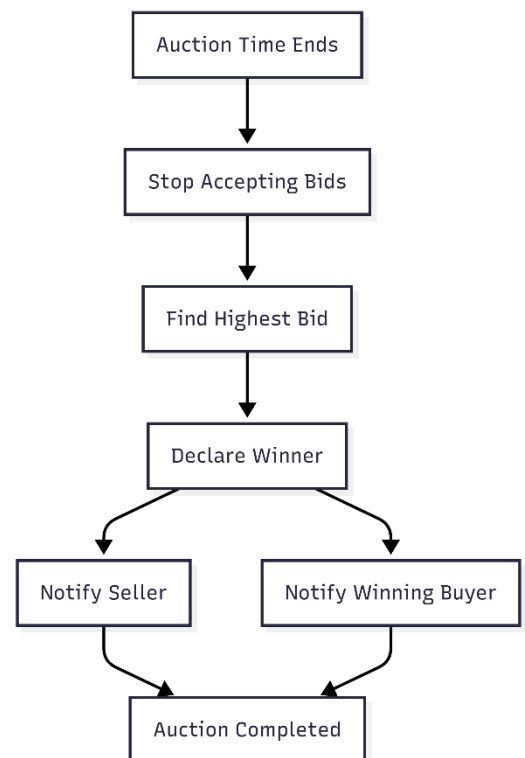
- The buyer enters a bid amount.
- The system checks whether the bid is higher than the current highest bid.
- If the bid is valid, it is accepted and saved.
- The highest bid is updated.
- If the bid is lower than the current highest bid, the system rejects it.
- The buyer can submit a higher bid.



4.5 Auction Result

Explanation

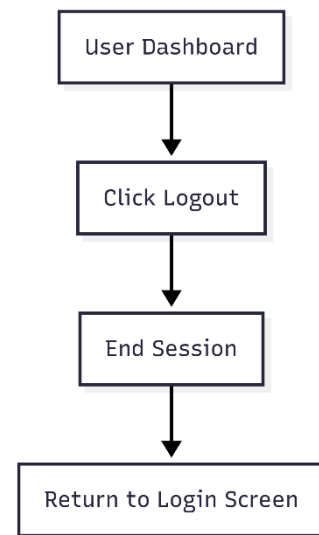
- The auction automatically closes when the end time is reached.
- The system stops accepting new bids.
- The highest valid bid is identified.
- The highest bidder is declared the winner.
- The seller and the winning buyer are notified.
- The auction status is updated to **Completed**.



4.6 Logout

Explanation

- The user clicks the **Logout** button.
- The system ends the active session.
- User information is cleared from the session.
- The user is redirected to the login page.
- The application is ready for the next login.



5. Application Protocol

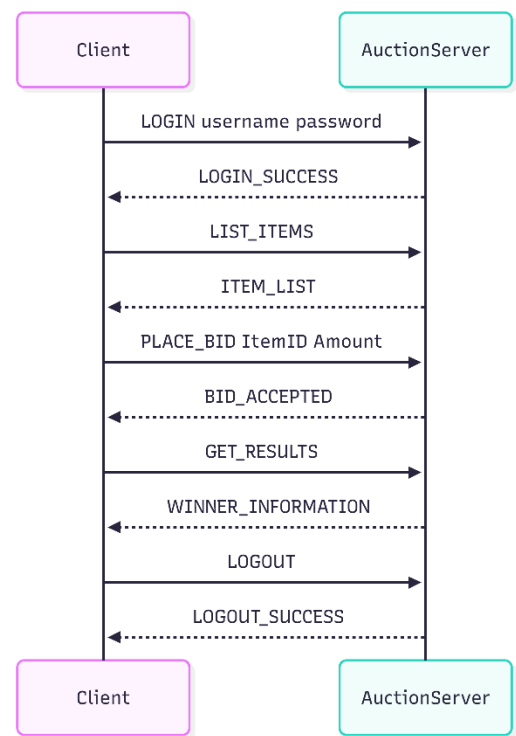
This is the networking part of your assignment. Since it's a **Client-Server project**, your professor expects you to design the protocol (the messages exchanged between the client and server).

5.1 Protocol Overview

Explanation

The Online Auction System uses a custom **TCP-based application protocol** for communication between the Client and the Auction Server. Every request sent by the client follows a predefined format, and the server responds with an appropriate status message. This protocol ensures reliable communication and supports all auction operations.

5.2 Protocol Workflow



5.3 Protocol Commands

Command	Purpose	Example
LOGIN	User authentication	LOGIN anil 1234
LOGOUT	End user session	LOGOUT
LIST_ITEMS	View available auction items	LIST_ITEMS
SEARCH_ITEM	Search by keyword	SEARCH_ITEM Laptop
VIEW_ITEM	View item details	VIEW_ITEM 101
ADD_ITEM	Seller uploads item	ADD_ITEM Laptop 500 24h
PLACE_BID	Buyer places bid	PLACE_BID 101 650
GET_RESULTS	View auction result	GET_RESULTS

5.4 Request Format

[COMMAND PARAMETER1 PARAMETER2 PARAMETER3]

Examples:

LOGIN anil 12345

ADD_ITEM Laptop 500 24h

PLACE_BID 101 700

SEARCH_ITEM Mobile

LOGOUT

5.5 Response Format

[STATUS MESSAGE]

Example:

LOGIN_SUCCESS

LOGIN_FAILED

ITEM_ADDED

BID_ACCEPTED

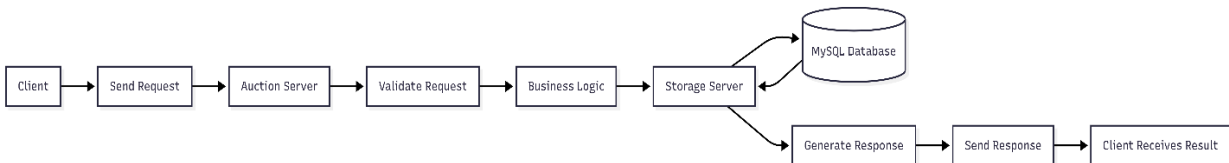
BID_REJECTED

ITEM_NOT_FOUND

LOGOUT_SUCCESS

SERVER_ERROR

5.6 Communication Flow



5.7 Error Messages

Error	Description
LOGIN_FAILED	Invalid username or password
ITEM_NOT_FOUND	Requested item does not exist
BID_TOO_LOW	Bid is lower than the current highest bid
AUCTION_CLOSED	Auction has already ended
INVALID_REQUEST	Incorrect command format
SERVER_ERROR	Internal server error

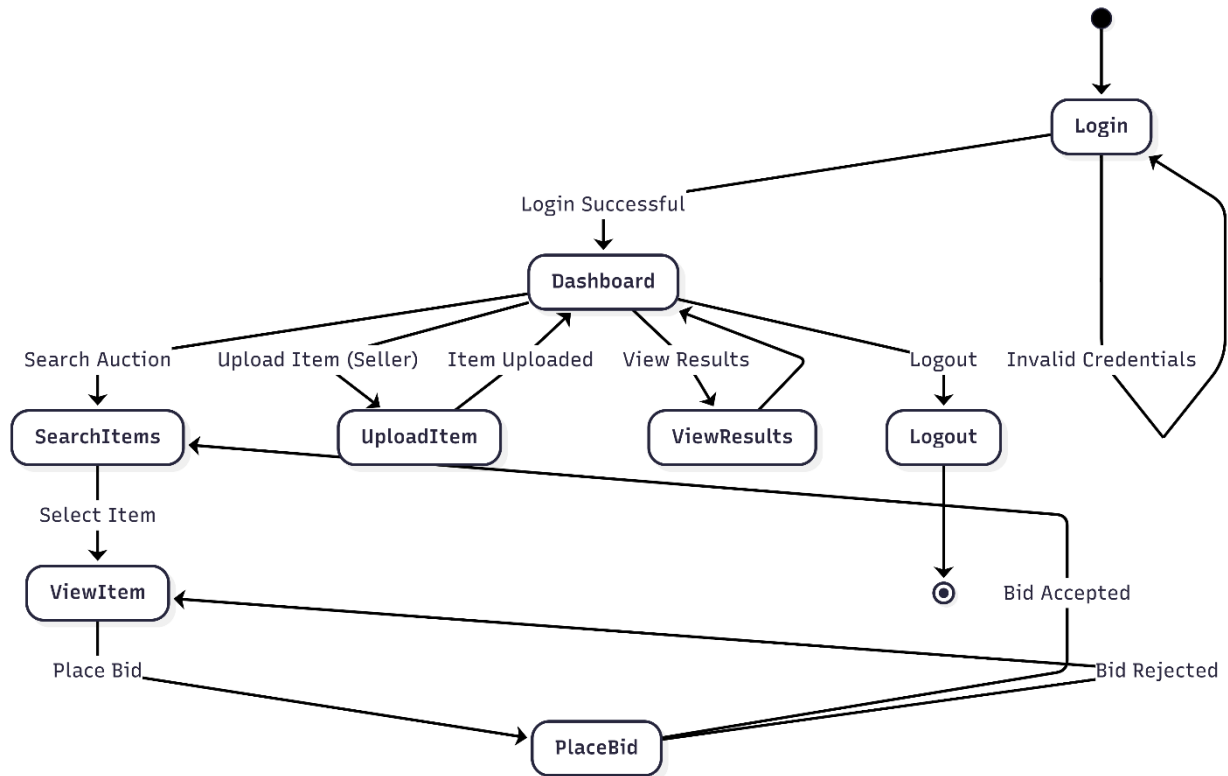
5.8 Example Communication Session

- Client → LOGIN anil 1234
- Server → LOGIN_SUCCESS
- Client → LIST_ITEMS
- Server → ITEM_LIST
- 101 Laptop ₹50,000
- 102 Phone ₹20,000
- Client → PLACE_BID 101 55000
- Server → BID_ACCEPTED
- Client → GET_RESULTS
- Server → YOU_ARE_HIGHEST_BIDDER
- Client → LOGOUT
- Server → LOGOUT_SUCCESS

6. Client State Chart

This diagram shows **the different states a client goes through while using the Online Auction System.**

6.1 Client State Diagram



Explanation

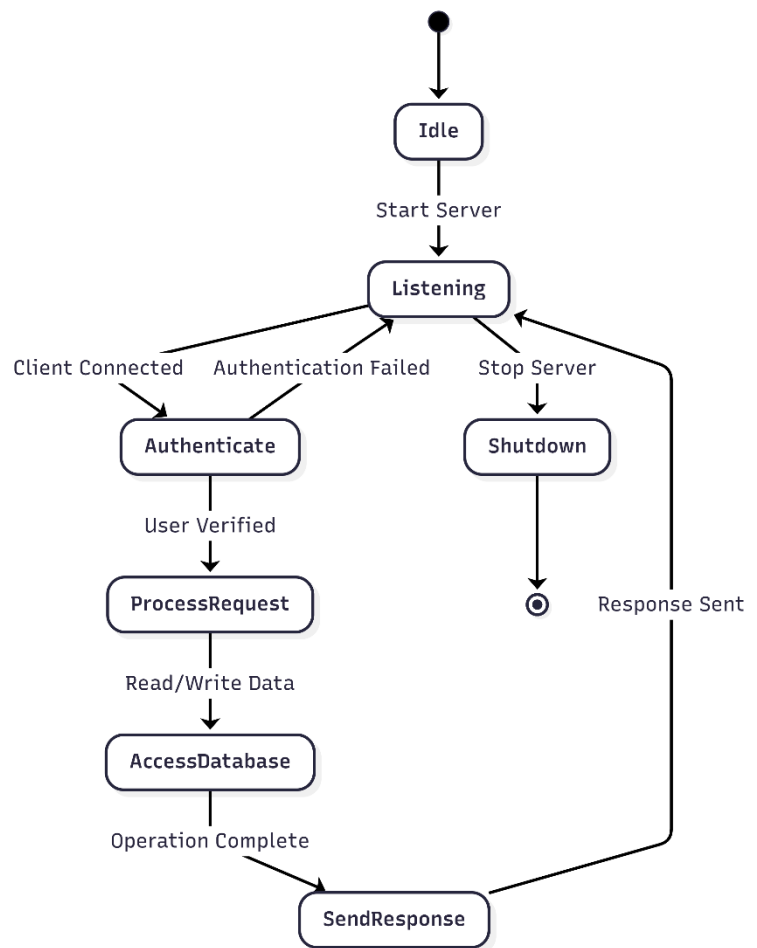
- The client starts at the Login screen.
- After successful authentication, the Dashboard is displayed.
- Buyers can search for items, view item details, and place bids.
- Sellers can upload new auction items.
- Users can check auction results at any time.
- When the user finishes, they log out and the session ends.

7. Server State Chart

The server continuously waits for client requests, processes them, and sends responses.

Explanation

- The server starts in the **Idle** state.
- It begins listening for client connections.
- When a client connects, the server authenticates the user.
- After successful authentication, it processes the client's request.
- If data is needed, it accesses the database.
- The server sends the response back to the client.
- The server returns to the listening state, ready to handle the next request.
- The process ends only when the server is shut down.



8. Example Communication Session

This is the final required section.

Explanation

- The client logs into the system using valid credentials.
- The Auction Server verifies the user with the database.
- After login, the client requests the list of auction items.
- The server retrieves the item list and returns it to the client.
- The client places a bid on an item.
- The server validates and stores the bid in the database.
- A confirmation is sent back to the client.
- Finally, the client logs out and the session ends.

